Practical #6: Practice question on Maven + Docker Integration

Statement: “Containerizing a Java Web Application for Cloud Deployment”

You are working as a DevOps engineer in a startup. The development team has built a Java web application (WAR file) using Maven. Your task is to:

- Build the WAR using Maven
- Containerize the application using Docker
- Deploy it on a containerized environment (simulating cloud deployment)

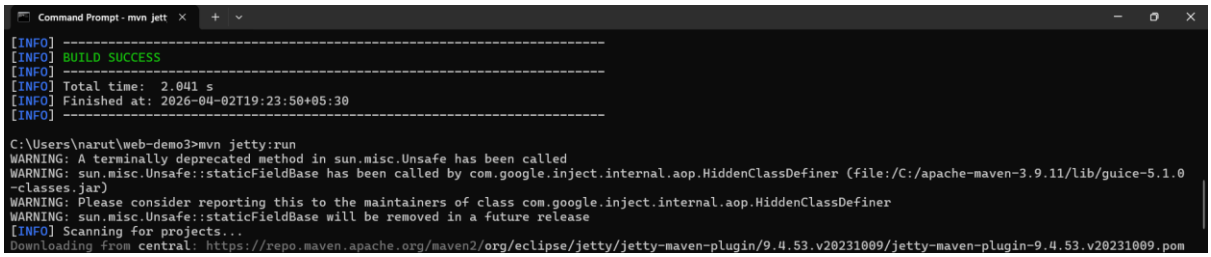
```
Command Prompt - mvn jett
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\narut>mvn archetype:generate -DgroupId=com.demo -DartifactId=web-demo3 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0
-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: C:\Users\narut
[INFO] Parameter: package, Value: com.demo
[INFO] Parameter: groupId, Value: com.demo
[INFO] Parameter: artifactId, Value: web-demo3
[INFO] Parameter: packageName, Value: com.demo
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\narut\web-demo3
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.379 s
[INFO] Finished at: 2026-04-02T19:19:56+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\narut>cd web-demo3
```

STEP-01: Create the template & get inside the folder

```
Command Prompt - mvn jett
C:\Users\narut\web-demo3>notepad pom.xml

C:\Users\narut\web-demo3>notepad dockerfile

C:\Users\narut\web-demo3>ren dockerfile.txt dockerfile

C:\Users\narut\web-demo3>dir
 Volume in drive C is Windows
 Volume Serial Number is 2461-2C7D

 Directory of C:\Users\narut\web-demo3

02-04-2026  19:22    <DIR>          .
02-04-2026  19:19    <DIR>          ..
02-04-2026  19:22                96 dockerfile
02-04-2026  19:21             1,302 pom.xml
02-04-2026  19:19    <DIR>          src
               2 File(s)          1,398 bytes
               3 Dir(s)   9,254,408,192 bytes free

C:\Users\narut\web-demo3>mvn clean install
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0
-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:web-demo3 >------------------------
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ web-demo3 ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ web-demo3 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ web-demo3 ---
[INFO] No sources to compile
```

STEP-02: Edit the pom.xml and make a docker file

STEP-03: mvn clean install



STEP-04: Run the container

STEP-05: Open your browser and visit:

http://localhost:8080/